

Proveedores de servicio

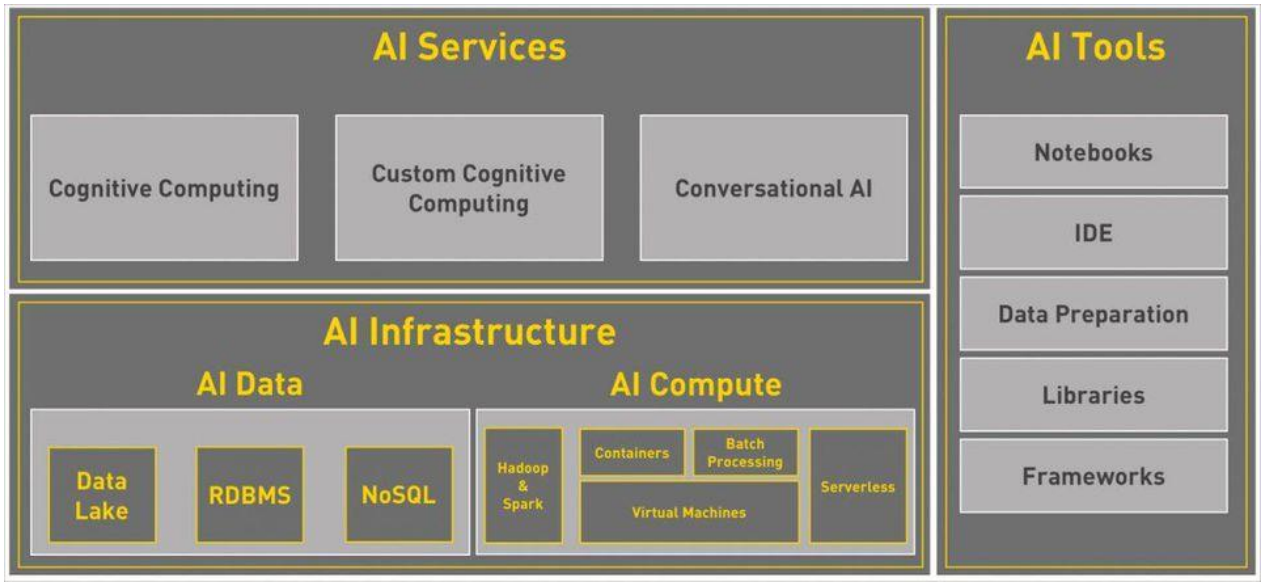
Rafael Pastor Vargas
SCC-ETSI Informática-UNED

PROVEEDORES DE SERVICIO

- Plataformas/frameworks locales (instalables/usables)
  - Scikit-Learn, TensorFlow, Deeplearning4J, PIO, H2O, Hadoop/Spark, etc.
- MLaaS (Machine Learning as a Service)
  - Basadas en Cloud
  - ABM, BigML, Google ML, AmazonML , AzureML , Watson IBM, etc.

https://jaxenter.com/top-5-machine-learning-service-providers-141275.html

https://analyticsindiamag.com/10-machine-learning-service-mlaas-tools-data-scientists/



Source: Janakiram MSV
AI Stack

- https://www.quora.com/What-are-the-best-machine-learning-as-a-service-MLaaS-companies-and-startups

MLAAS

AmazonML: Amazon Machine Learning: use cases and a real example

AzureML: Azure Machine Learning: simplified predictive analytics

BigML: BigML: Machine Learning made easy

Datoin: Build Intelligent Apps.

ParallelDots: ParallelDots

DeepAI: Enterprise machine learning as a service

Deep Cognition: Deep Learning Studio simplifies AI development and deployment

Ersatz: Hosted deep learning with GPUs

ForecastThis: automated predictive modeling solutions

Google Prediction API: Google Prediction API: a Machine Learning black box for devs

MLJAR: Platform for building Machine Learning models

Natero: Customer Success platform that predicts churn and up-sell opportunities

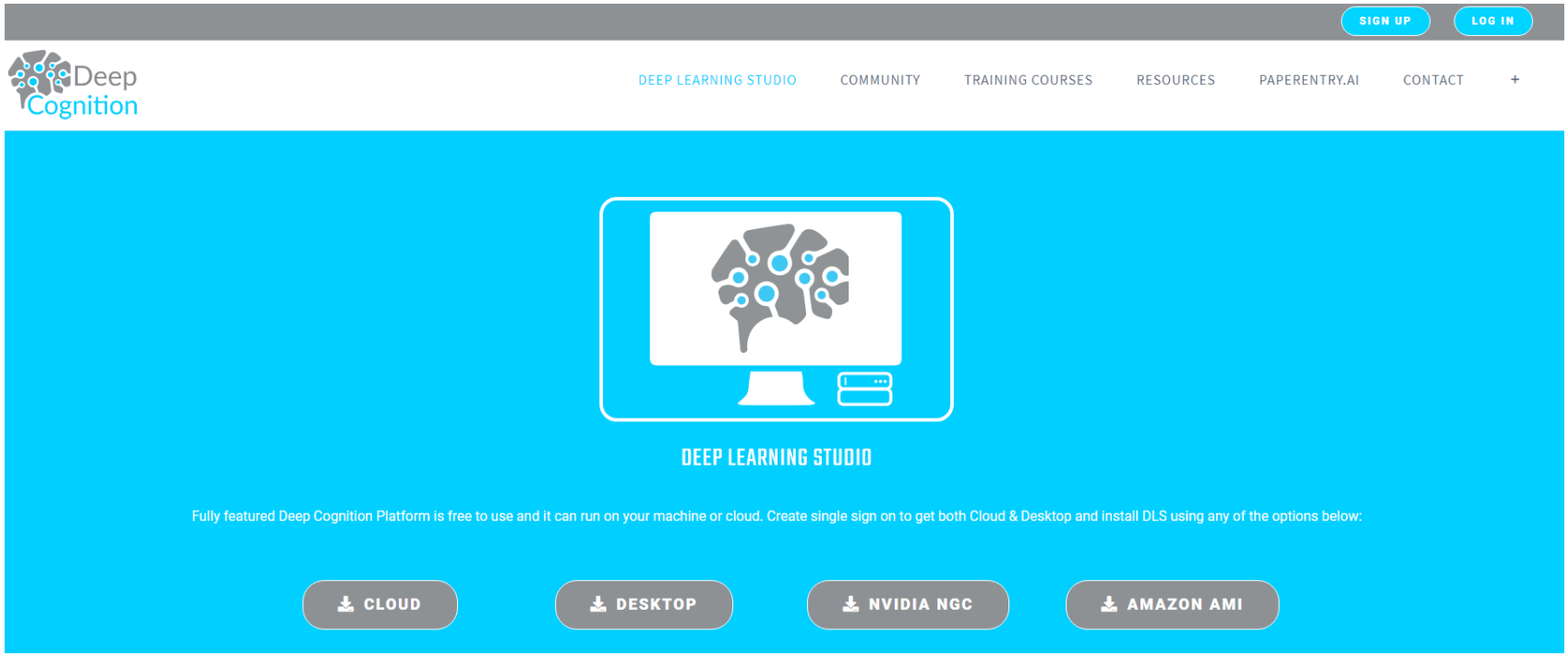
DataRobot: Machine Learning Software - DataRobot

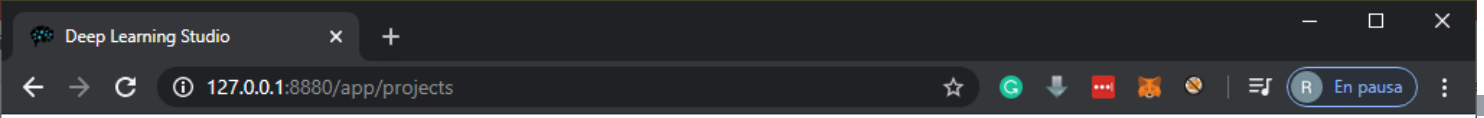
Vize AI: Custom image recognition API

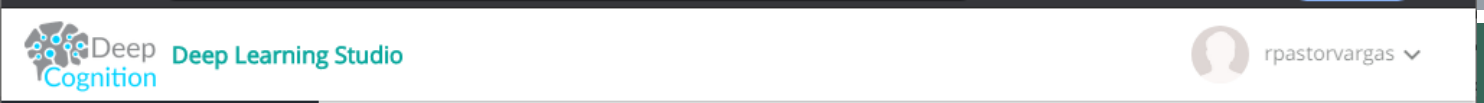
Watson Analytics: Guided and automated analytics from the cloud.

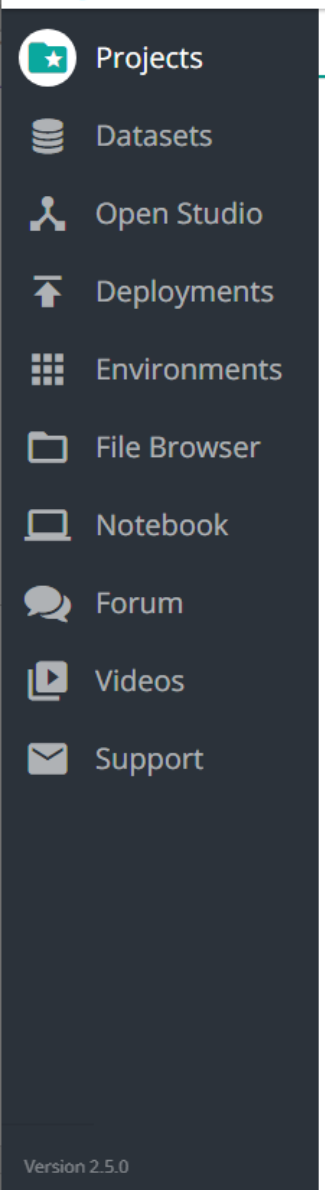
Wise: Machine Learning for Customer Success:

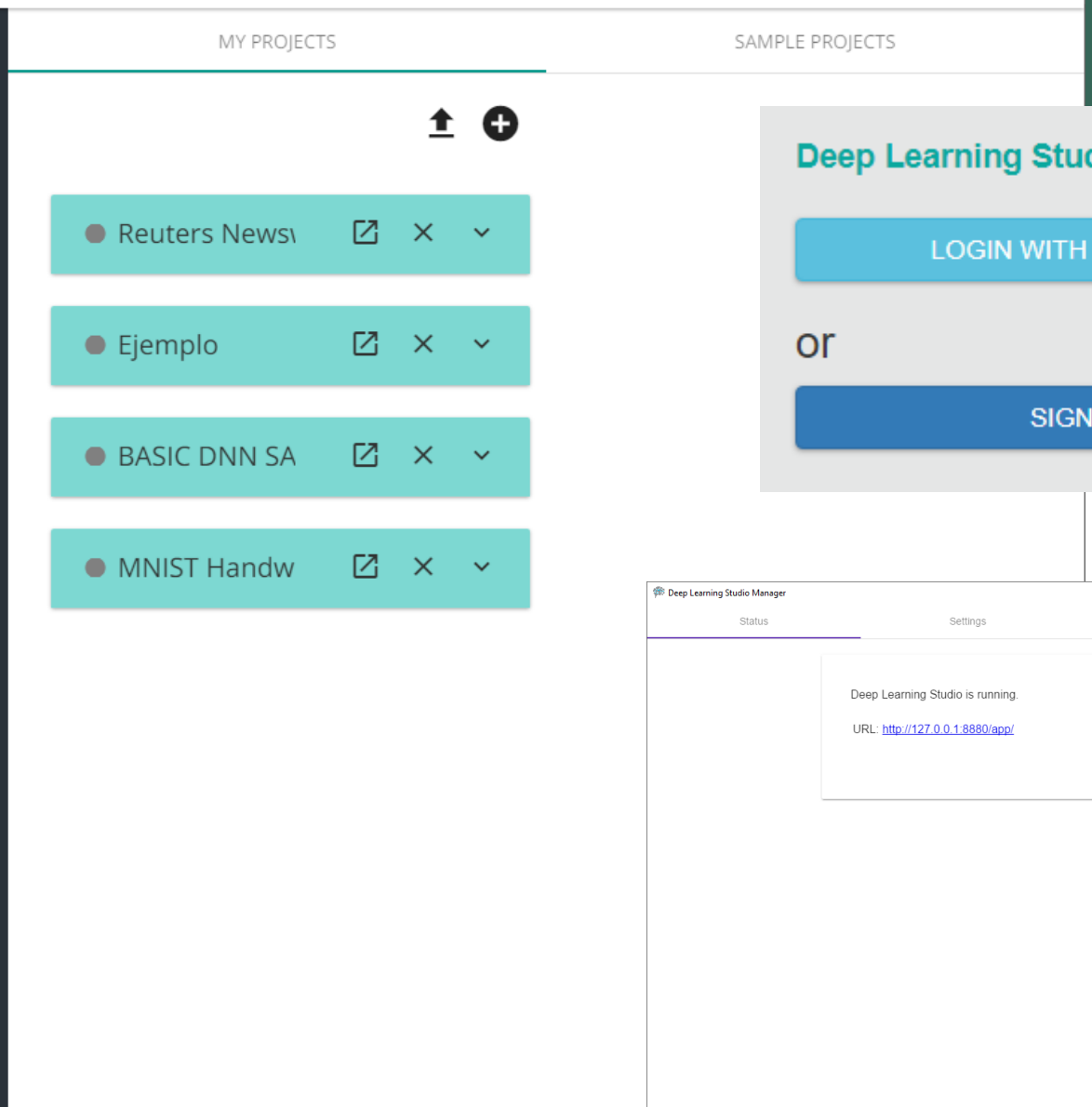
DEEPCOGNITION



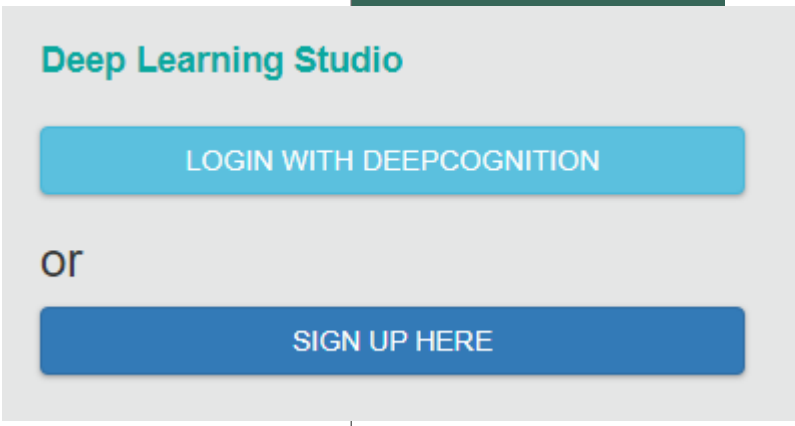

Deep Learning Studio
127.0.0.1:8880/app/projects
En pausa

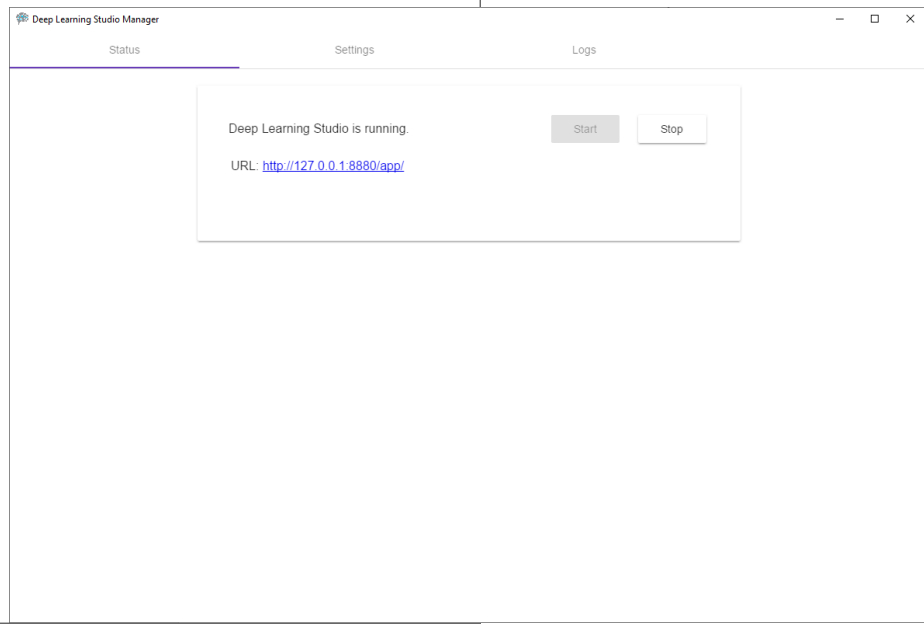

Deep Cognition
Deep Learning Studio
rpastorvargas

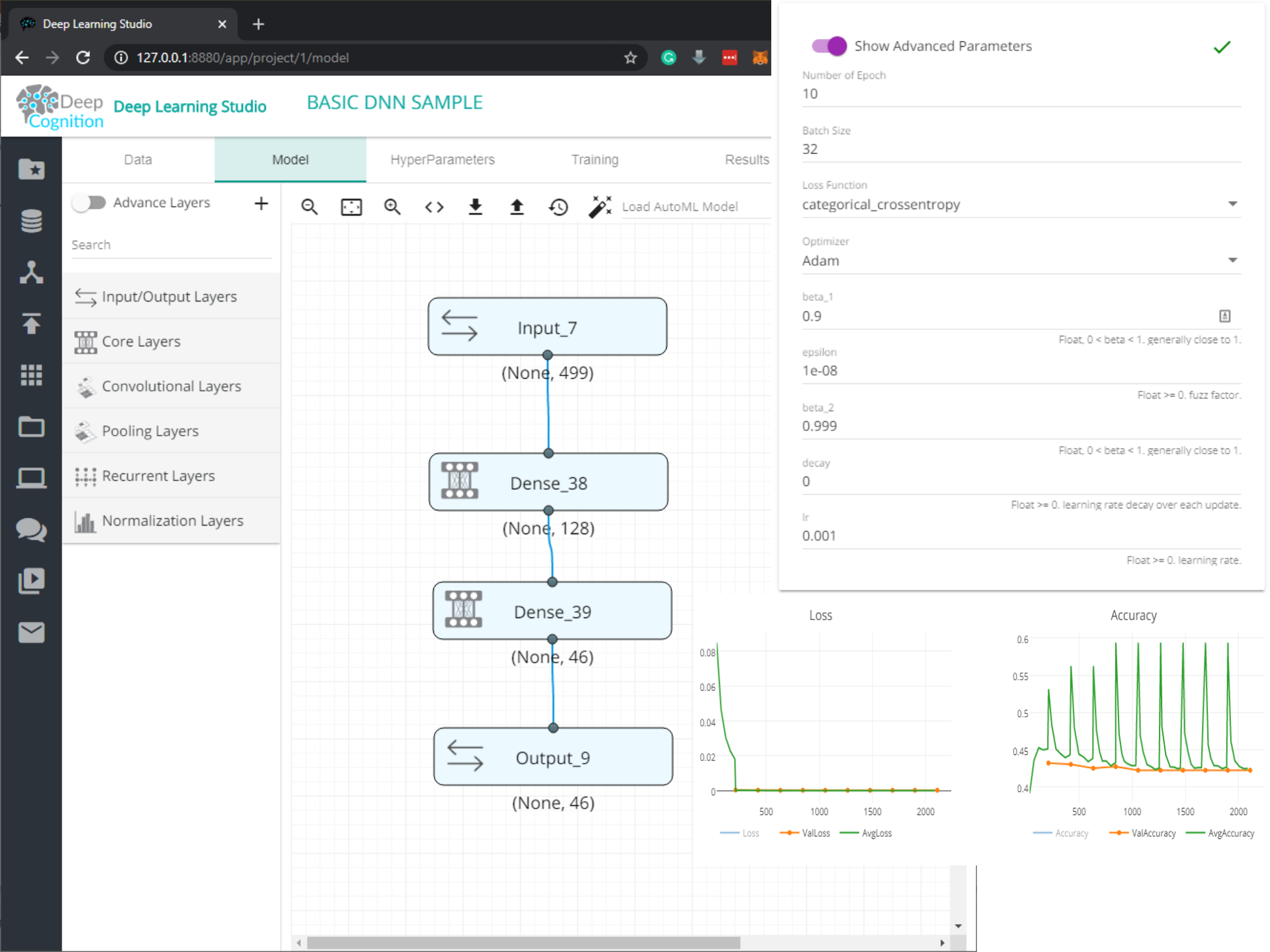

Projects
Datasets
Open Studio
Deployments
Environments
File Browser
Notebook
Forum
Videos
Support
Version 2.5.0
Copyright © 2020 Deep Cognition


MY PROJECTS
SAMPLE PROJECTS
Reuters News\
Ejemplo
BASIC DNN SA
MNIST Handw




Deep Learning Studio
LOGIN WITH DEEPCOGNITION
or
SIGN UP HERE


Deep Learning Studio Manager
Status
Settings
Logs
Deep Learning Studio is running.
Start
Stop
URL: http://127.0.0.1:8880/app/


Deep Learning Studio
127.0.0.1:8880/app/project/1/model
Deep Cognition
Deep Learning Studio
BASIC DNN SAMPLE
Data
Model
HyperParameters
Training
Results
Advance Layers
Search
Input/Output Layers
Core Layers
Convolutional Layers
Pooling Layers
Recurrent Layers
Normalization Layers
Load AutoML Model
Input_7
(None, 499)
Dense_38
(None, 128)
Dense_39
(None, 46)
Output_9
(None, 46)
Show Advanced Parameters
Number of Epoch
10
Batch Size
32
Loss Function
categorical_crossentropy
Optimizer
Adam
beta_1
0.9
Float, 0 < beta < 1. generally close to 1.
epsilon
1e-08
Float >= 0. fuzz factor.
beta_2
0.999
Float, 0 < beta < 1. generally close to 1.
decay
0
Float >= 0. learning rate decay over each update.
lr
0.001
Float >= 0. learning rate.
Loss
0.08
0.06
0.04
0.02
0
500
1000
1500
2000
Loss
ValLoss
AvgLoss
Accuracy
0.6
0.55
0.5
0.45
0.4
500
1000
1500
2000
Accuracy
ValAccuracy
AvgAccuracy

H2O.AI

H_2O

H2O works with R, Python, Scala on Hadoop/Yarn, Spark or your laptop

Nightly Bleeding Edge

Latest Stable Release

Verify this release using checksums.

H_2O is licensed under the Apache License, Version 2.0

Sparkling Water

Sparkling Water combines the fast, scalable machine learning algorithms of H2O with the capabilities of Spark

Sparkling Water 2.2

Sparkling Water 2.1

Sparkling Water 2.0

Nightly Bleeding Edge

Sparkling Water is licensed under the Apache License, Version 2.0

Driverless AI

Automatic Machine Learning for the Enterprise. Driverless AI automates feature engineering, model building, visualization and interpretability.

Download

Don't have a license key?

Sign Up Here

Driverless AI is a commercially licensed product.

Deep Water

Deep Water is H2O Deep Learning with GPU support for TensorFlow, MXNet, and Caffe

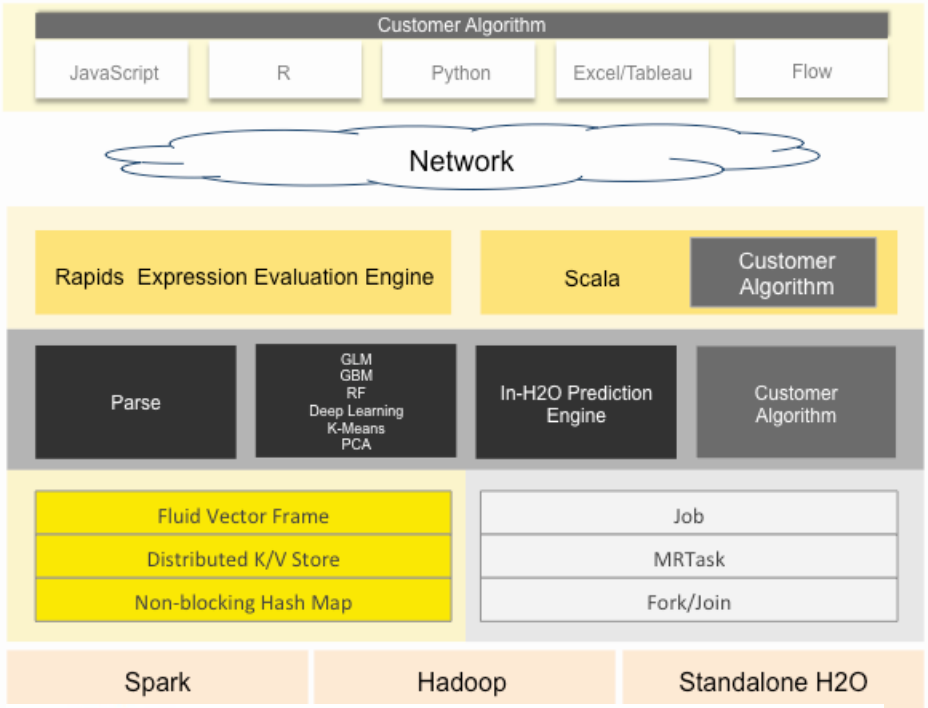
Try Deep Water

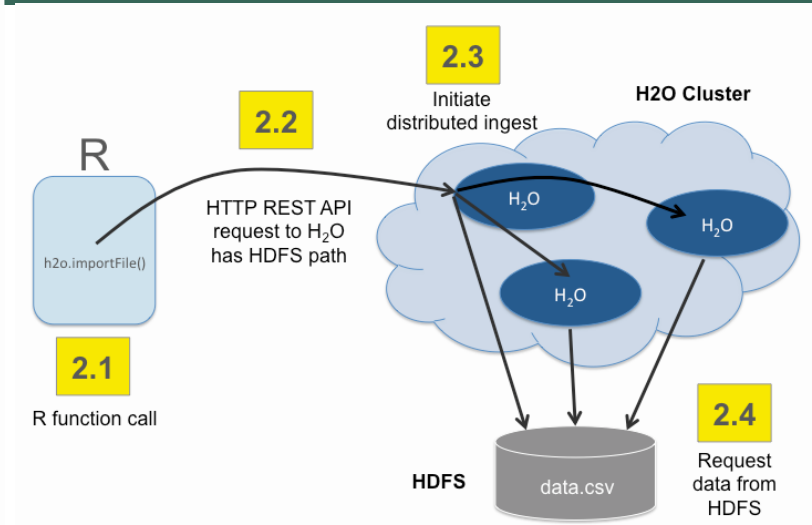
Deep Water is licensed under the Apache License, Version 2.0

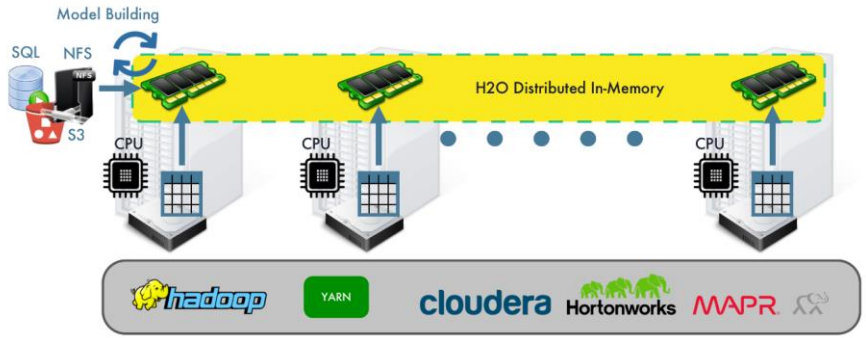
H_2O4GPU

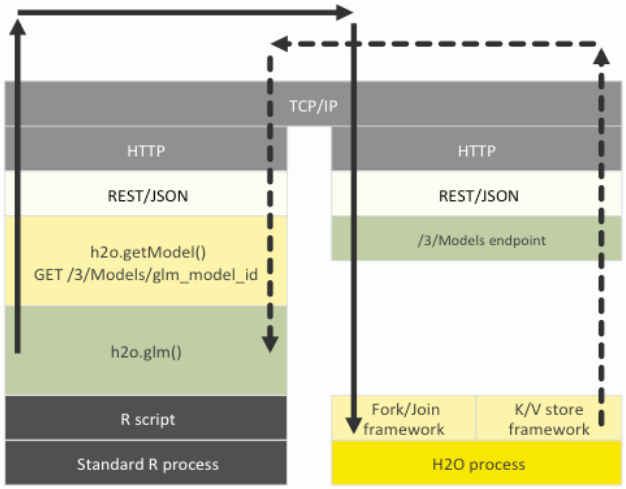
- http://docs.h2o.ai/h2o/latest-stable/h2o-docs/architecture.html

ARQUITECTURA DE H2O



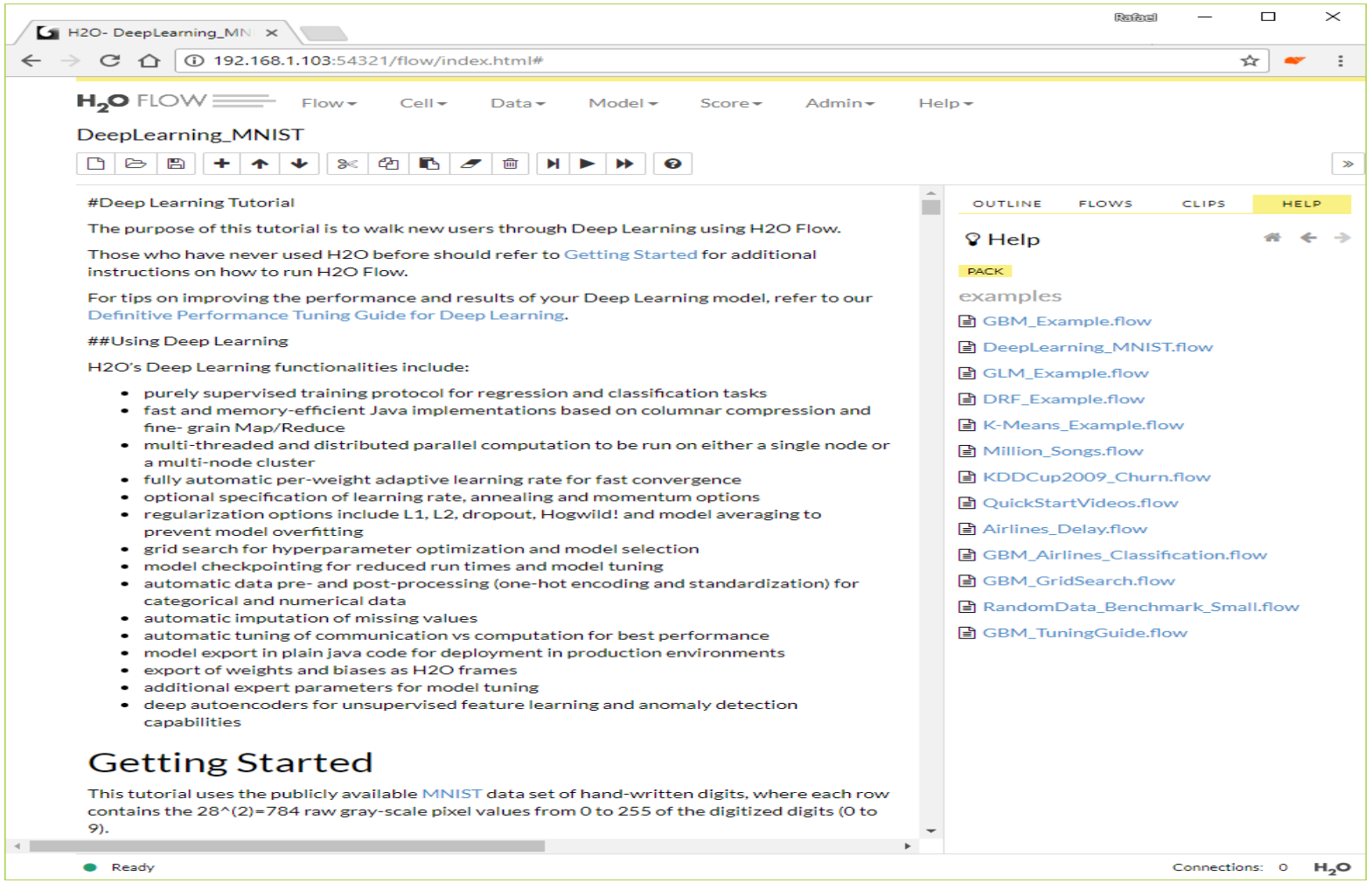








DEMO DEEP LEARNING Y H20 WEB FLOW UI



https://www.h2o.ai/blog/a-deep-dive-into-h2os-automl/

AUTO ML

Example

```
import h2o
from h2o.automl import H2OAutoML
h2o.init()

train = h2o.import_file("train.csv")

aml = H2OAutoML(max_runtime_secs = 600)
aml.train(y = "response_colname",
          training_frame = train)

lb = aml.leaderboard
```

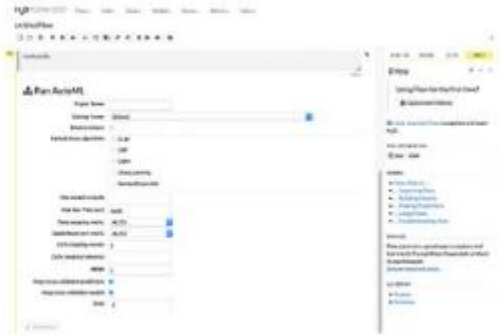
Example

```
library(h2o)
h2o.init()

train <- h2o.importFile("train.csv")

aml <- h2o.automl(y = "response_colname",
                  training_frame = train,
                  max_runtime_secs = 600)

lb <- aml@leaderboard
```



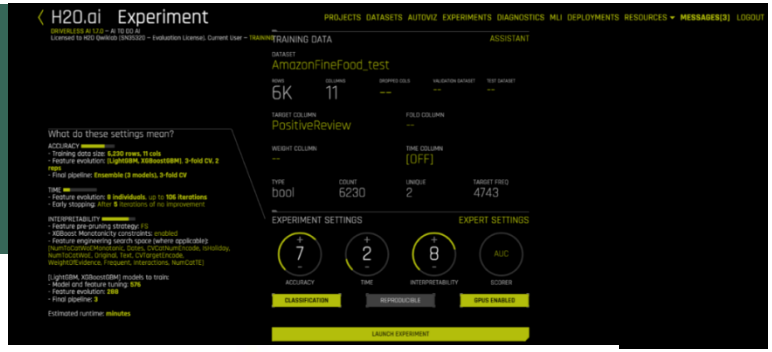
H2O AutoML in Python | H2O AutoML in R | H2O AutoML in Flow GUI

Load the Training & Test dataset → Specify the Response and Predictor variables. → Run AutoML specifying the stopping criterion → View the leaderboard → Explore the ensemble composition. → Save the Leader model

model_id	auc	logloss	mean_per_class_error	rmse	mse
StackedEnsemble_AllModels_AutoML_20191015_055901	0.950398	0.181045	0.147554	0.227075	0.051563
StackedEnsemble_BestOfFamily_AutoML_20191015_055901	0.950298	0.181333	0.151449	0.227149	0.0515968
XGBoost_1_AutoML_20191015_055901	0.949053	0.172296	0.15365	0.225251	0.050738
GBM_4_AutoML_20191015_055901	0.948941	0.17323	0.164059	0.225911	0.0510359
XGBoost_3_AutoML_20191015_055901	0.94797	0.1732	0.155337	0.226023	0.0510864
XGBoost_2_AutoML_20191015_055901	0.947655	0.17421	0.16391	0.226491	0.051298
GBM_3_AutoML_20191015_055901	0.947212	0.176003	0.171788	0.22788	0.0519294
GBM_5_AutoML_20191015_055901	0.947099	0.17653	0.163763	0.229752	0.0527858
GBM_2_AutoML_20191015_055901	0.944633	0.180209	0.159998	0.230551	0.0531539
GBM_1_AutoML_20191015_055901	0.941486	0.184062	0.164992	0.232392	0.0540058

https://www.h2o.ai/blog/custom-machine-learning-recipes-the-ingredients-for-success/

DRIVERLESS AI



1 Drag and Drop Data

2 Automatic Visualization

4 Automatic Model Optimization

✓ Driverless AI executes automation on your recipes

Feature engineering, model selection, hyper-parameter tuning, overfitting protection

5 Automatic Scoring Pipelines

✓ Driverless AI automates model scoring and deployment using your recipes

HDFS

SQL

Snowflake

Amazon S3

Google BigQuery

Azure Blog Storage

Modelling

X Y

Dataset

Automatic Model Optimization

Model Recipes
- i.i.d. data
- Time-series
- More on the way

Survival of the Fittest



Advanced Feature Engineering + Algorithm + Model Tuning

Model Documentation

Automatic Scoring Pipeline

Deploy Low-latency Scoring to Production

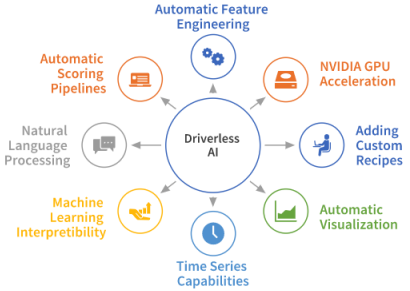
3 Bring Your Own Recipes

✓ Upload your own recipe(s)

Scorers

Algorithms

Transformations

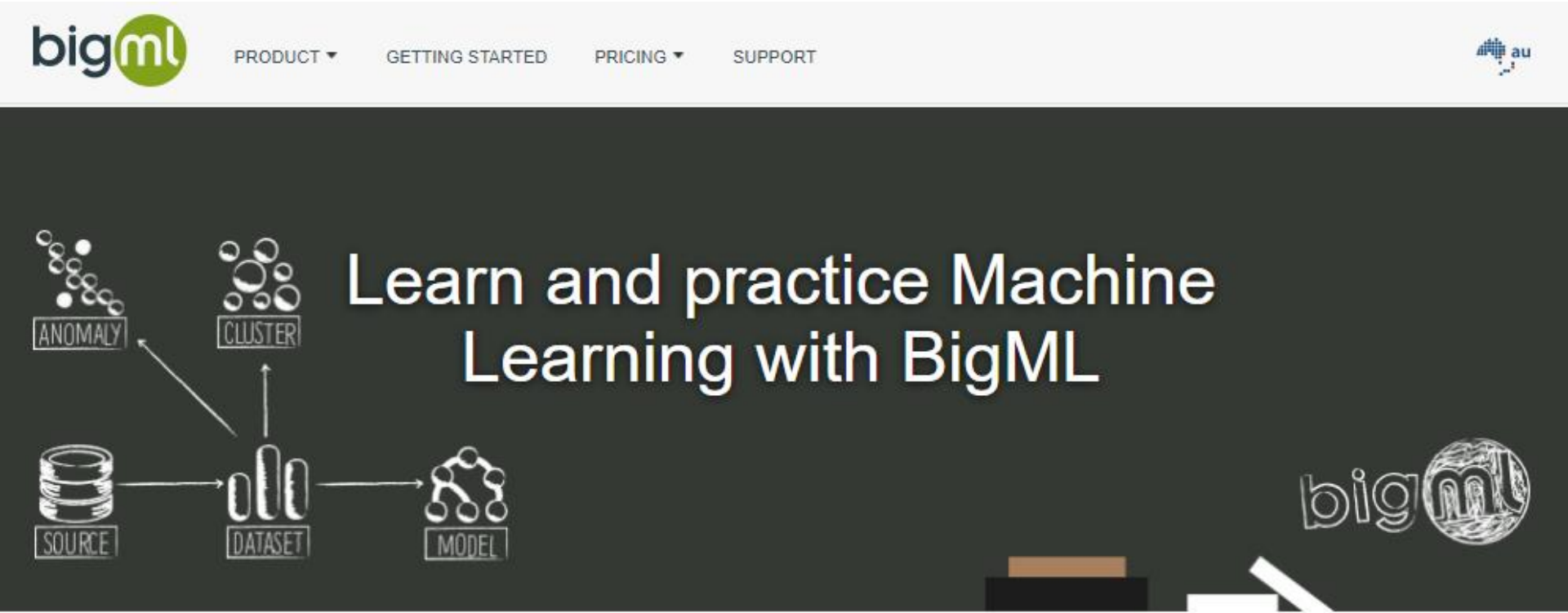


EJERCICIO

- Deep Learning en R
  - https://www.datacamp.com/community/tutorials/keras-r-deep-learning
  - https://www.r-bloggers.com/deep-learning-in-r-2/
  - https://towardsdatascience.com/how-to-implement-deep-learning-in-r-using-keras-and-tensorflow-82d135ae4889
- H2O
  - install.packages("h2o")
  - H2O_mnist_dl.R
  - http://docs.h2o.ai/h2o/latest-stable/h2o-docs/booklets/DeepLearningBooklet.pdf

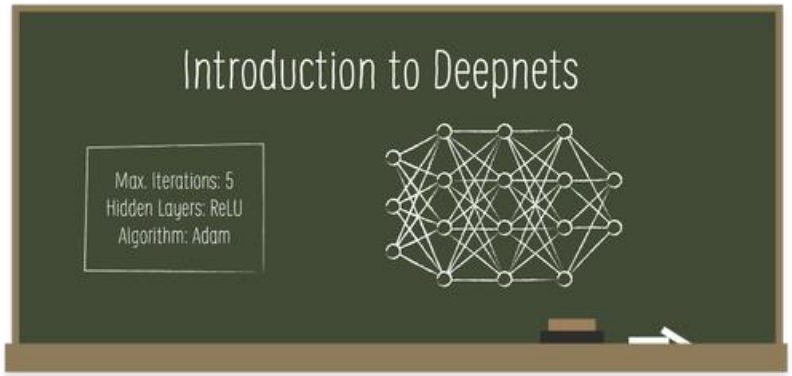
BIG ML

- Programable y escalable
- Tareas: Clasificación, Regresión, Análisis de Clusters, Detección de Anomalías, Descubrimiento de Asociación y Modelado.

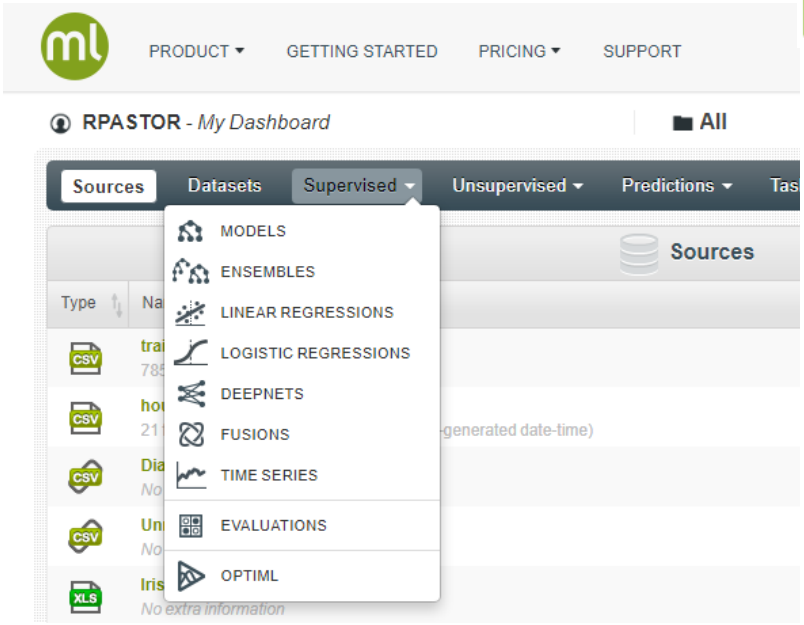


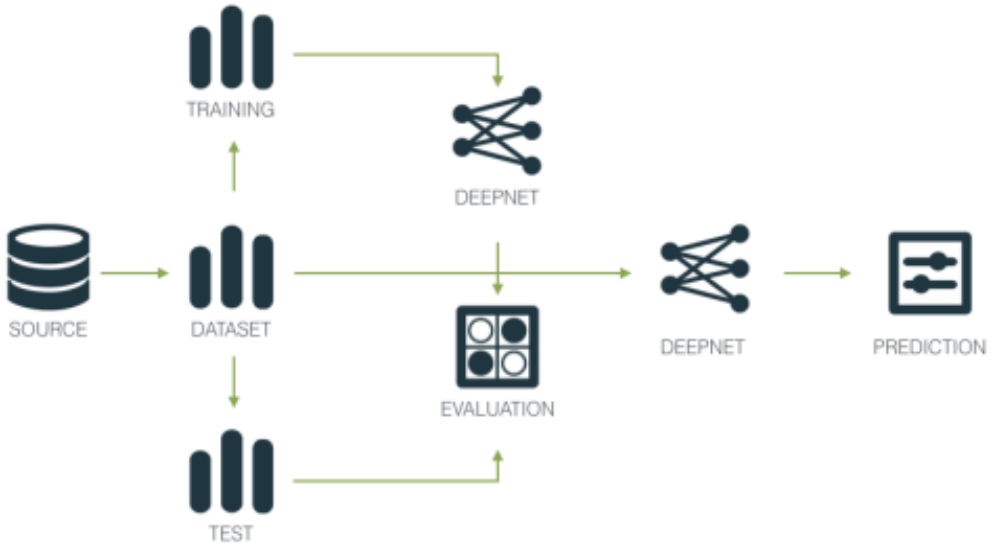
- https://blog.bigml.com/2017/09/26/introduction-to-deepnets/

BIG ML: DEEPNETS



ALGORITHM	PROs	CONs
Logistic Regression	• Handles smooth, multivariate functions for the most part • Trains real fast	• Falls short if decision boundary is nonlinear
Trees / Tree Ensembles	• Massive representational power	• Struggle with smooth functions (mitigated by ensembles to a large extent)
Deepnets	• Arbitrary representational power • Smooth, multivariate objectives don't present a problem	• Trial and error based search for the right neural network structure • Difficult to interpret

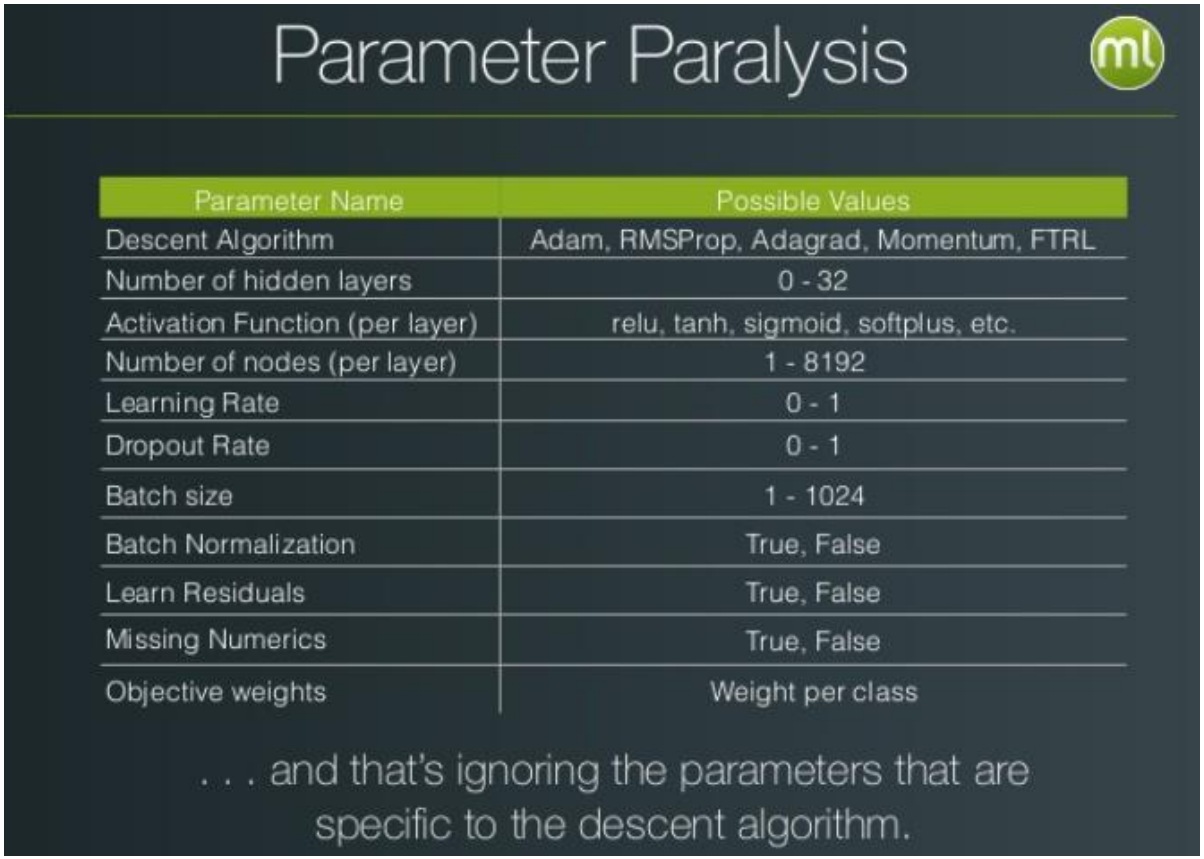




DEEPNETS (BIGML)

We trained 296,748 deep neural networks so you don't have to!

- https://www.slideshare.net/bigml/bigml-summer-2017-release
- No soporta:
  - CNN
  - RN



EJERCICIO

- https://blog.bigml.com/2017/09/29/creating-your-deepnets-with-the-bigml-dashboard/
- https://blog.bigml.com/2017/09/28/case-study-finding-higgs-bosons-with-deepnets/